Inteligencia Artificial

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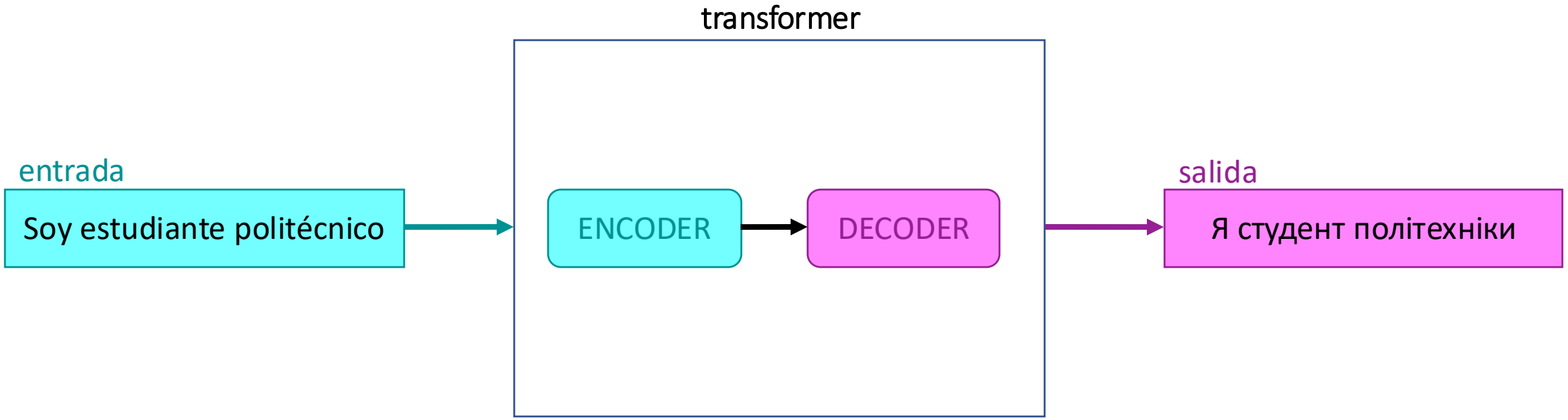
Facultad de Ingeniería en Electricidad y Computación – FIEC

Clase # 19 - PAO 2024 – II

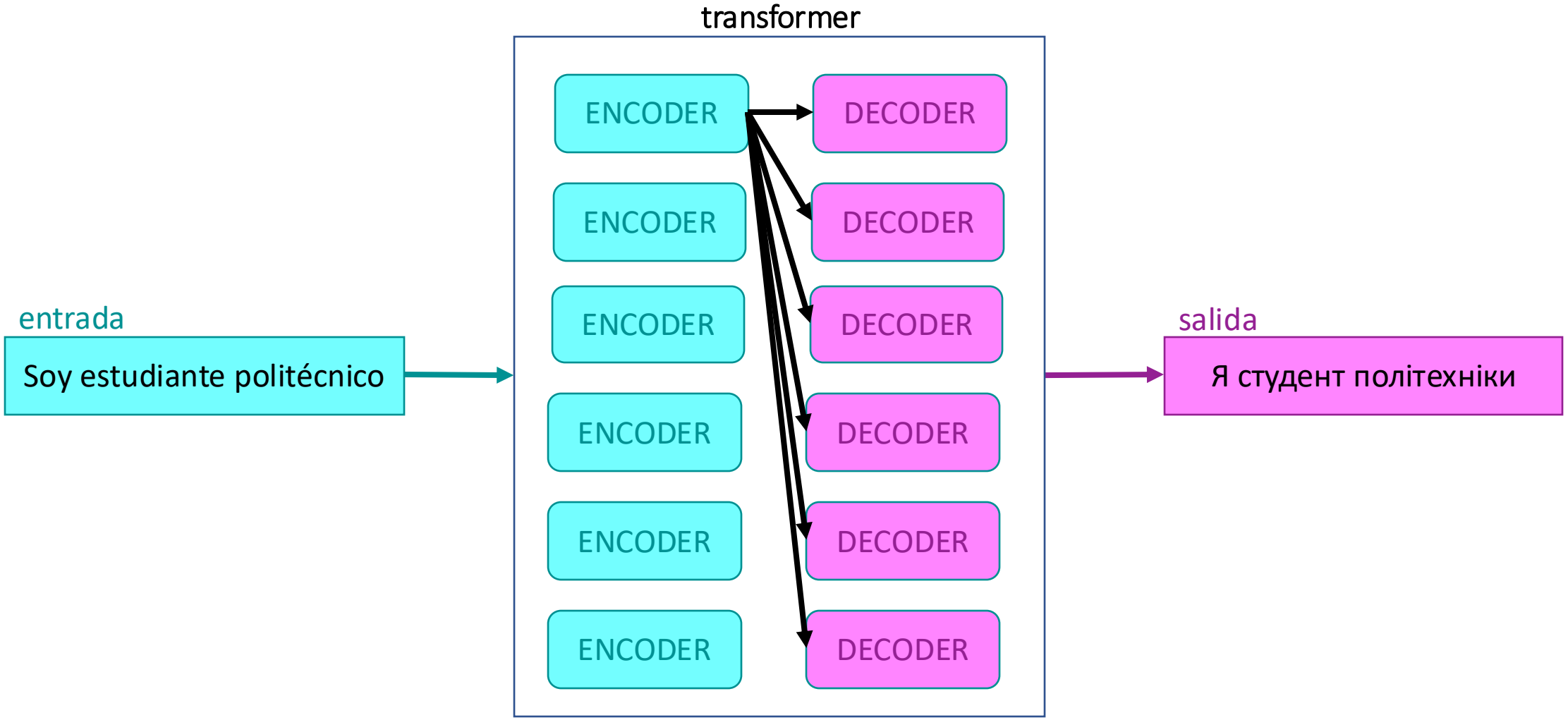
Agenda

1. Transformers
    - Qué son ?
    - Encoder – Decoder
    - Embeddings – positional embedding
    - Attention
    - Residuos y Normalización
    - Seq2Seq con attention
    - Salida

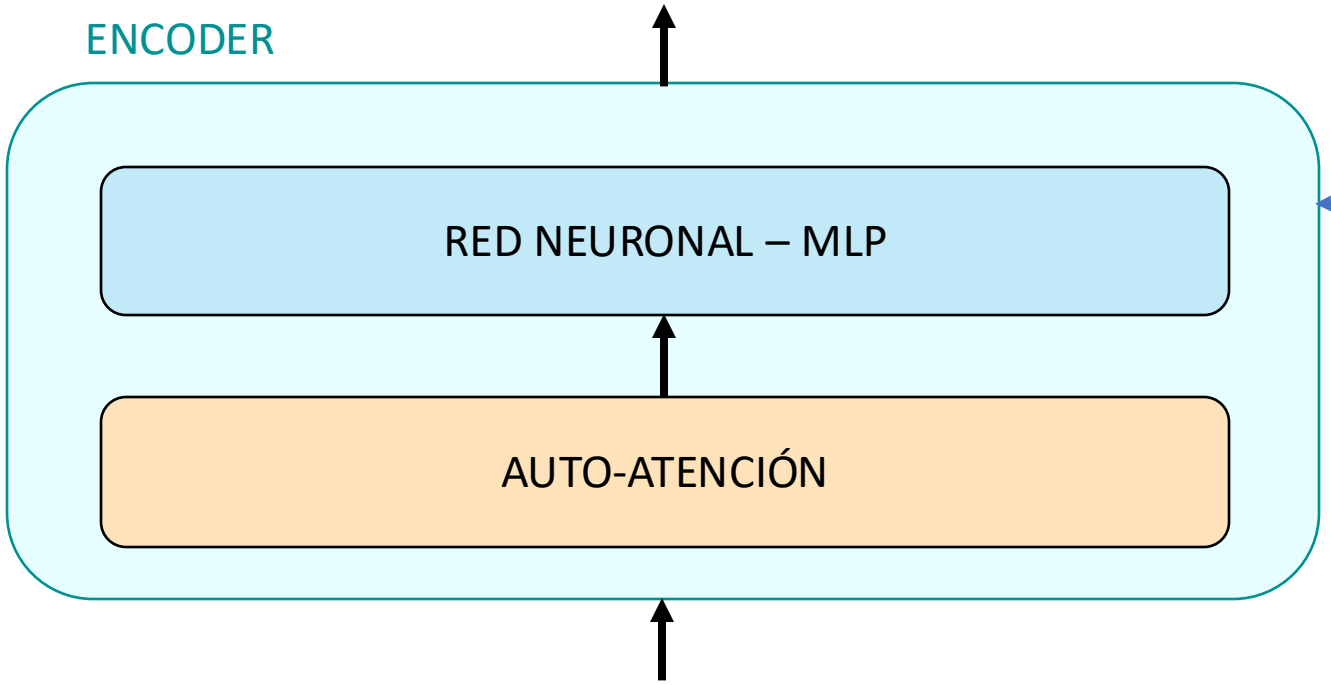
Transformers – Encoder – Decoder



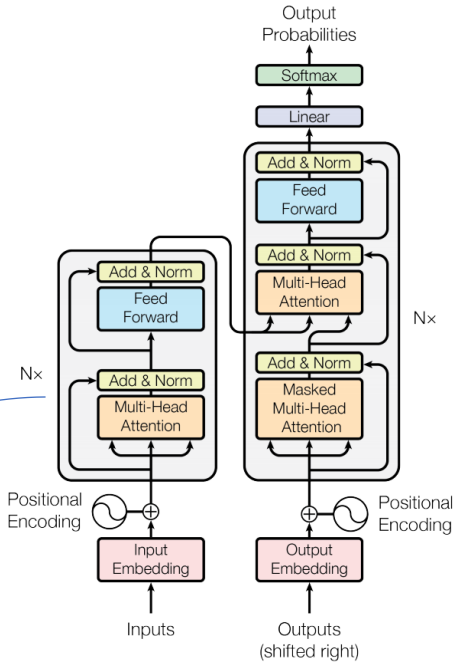
Transformers



Transformers



6 capas como esta



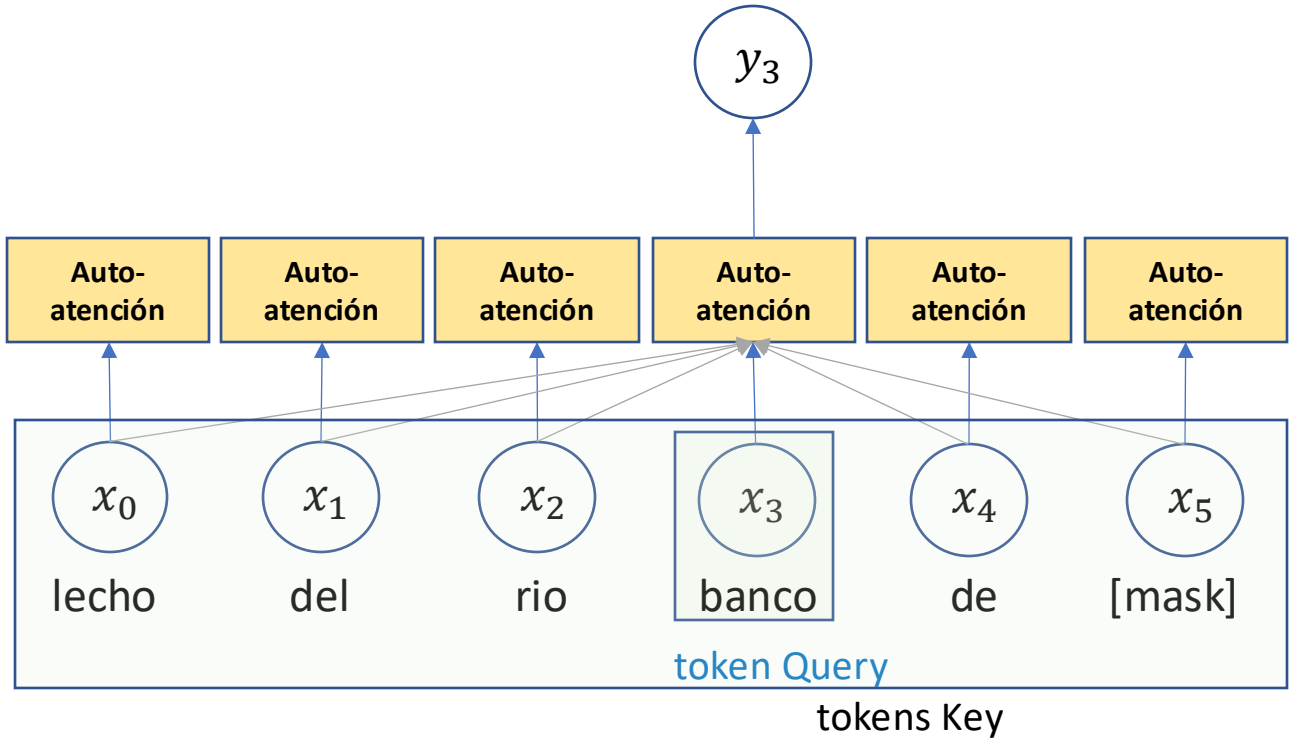
- Encoders son identicos, pero no comparten los pesos

Transformers

- Auto-atención

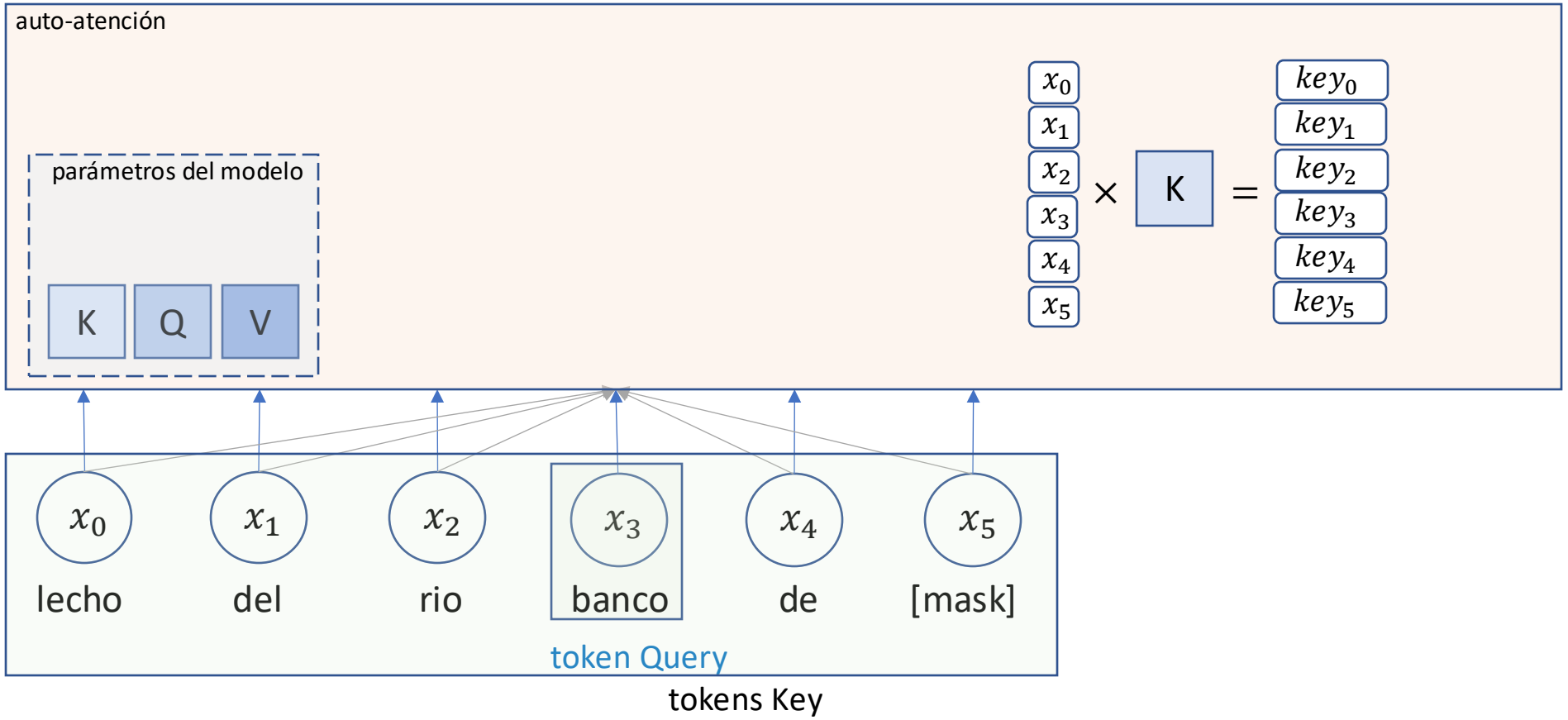
Embeddings Contextual Y_i

Embeddings no-Contextual X_i



Transformers

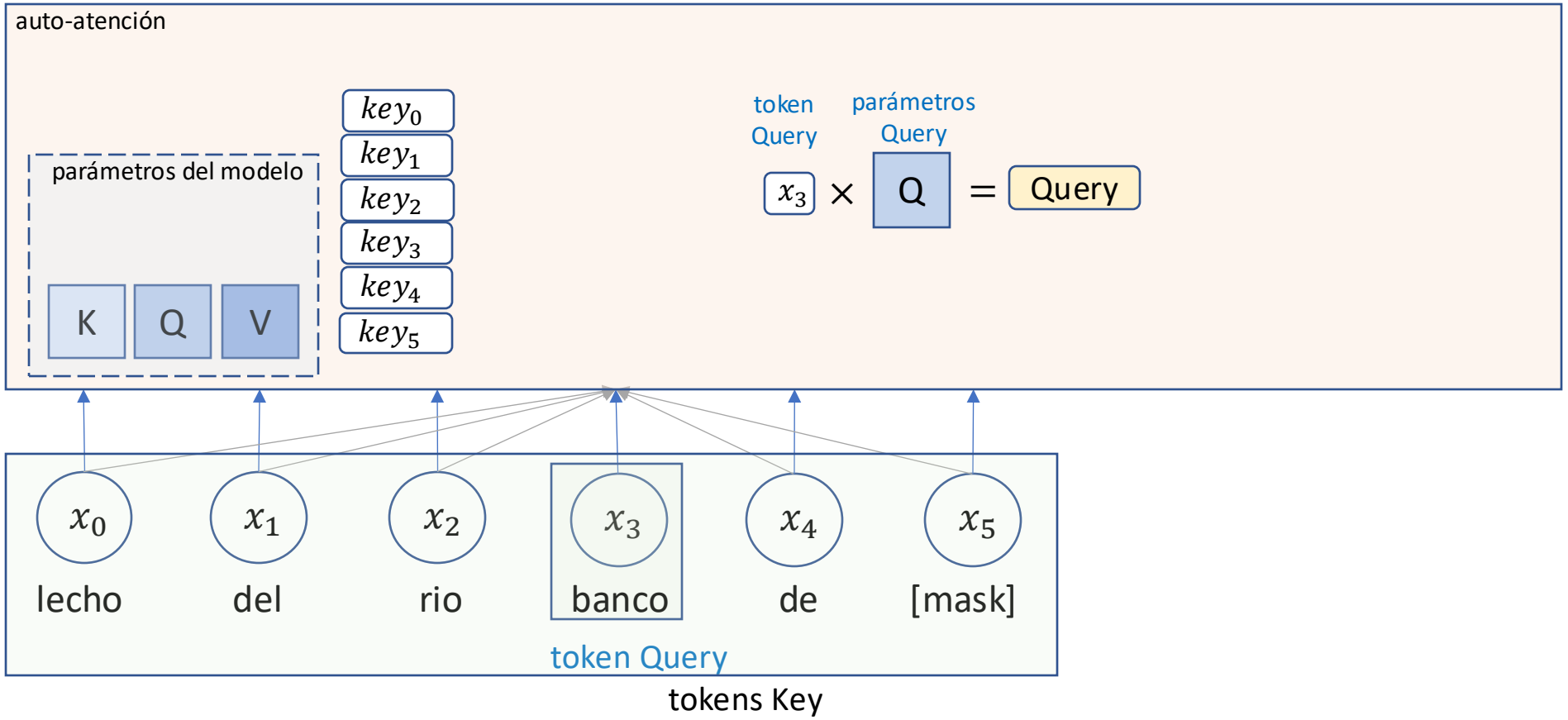
- Las Claves (Keys)



Embeddings no-Contextual X_i

Transformers

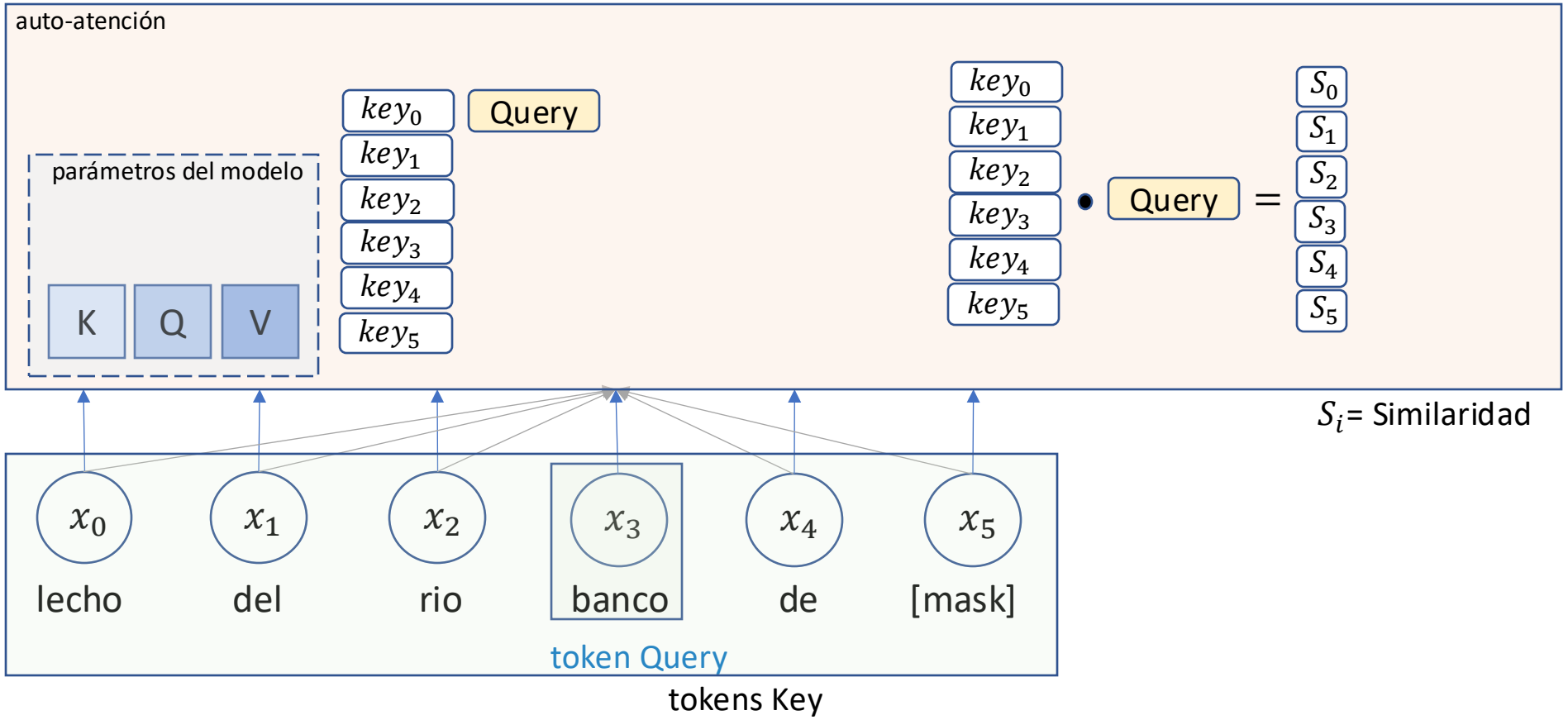
- Query



Embeddings no-Contextual X_i

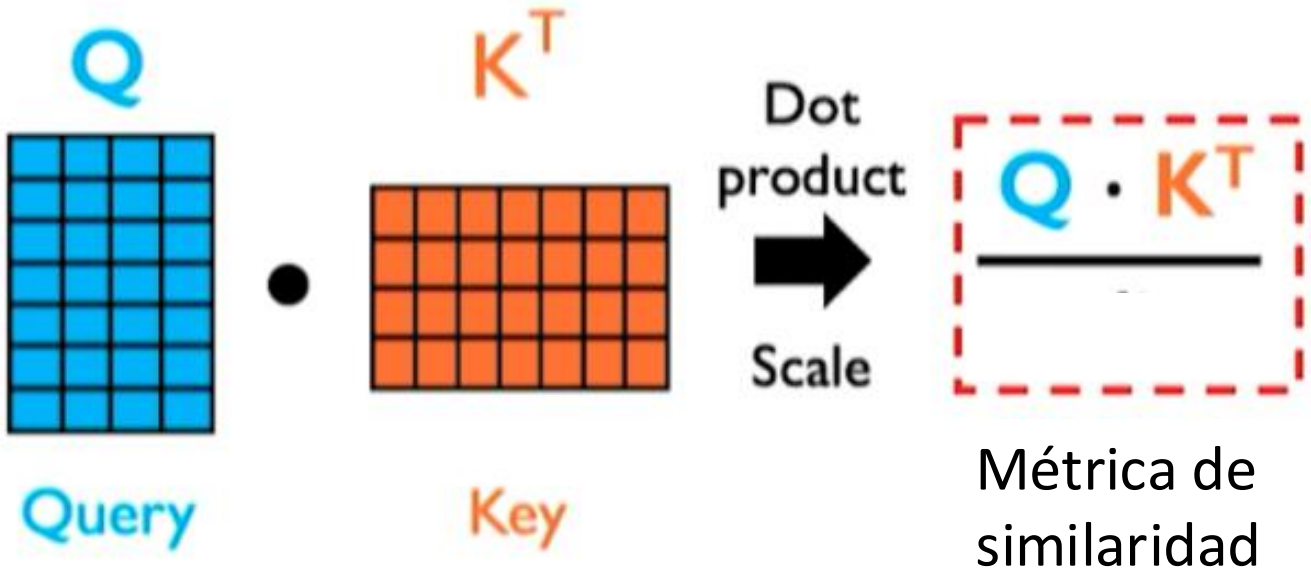
Transformers

- Las Valores



Embeddings no-Contextual X_i

Transformers

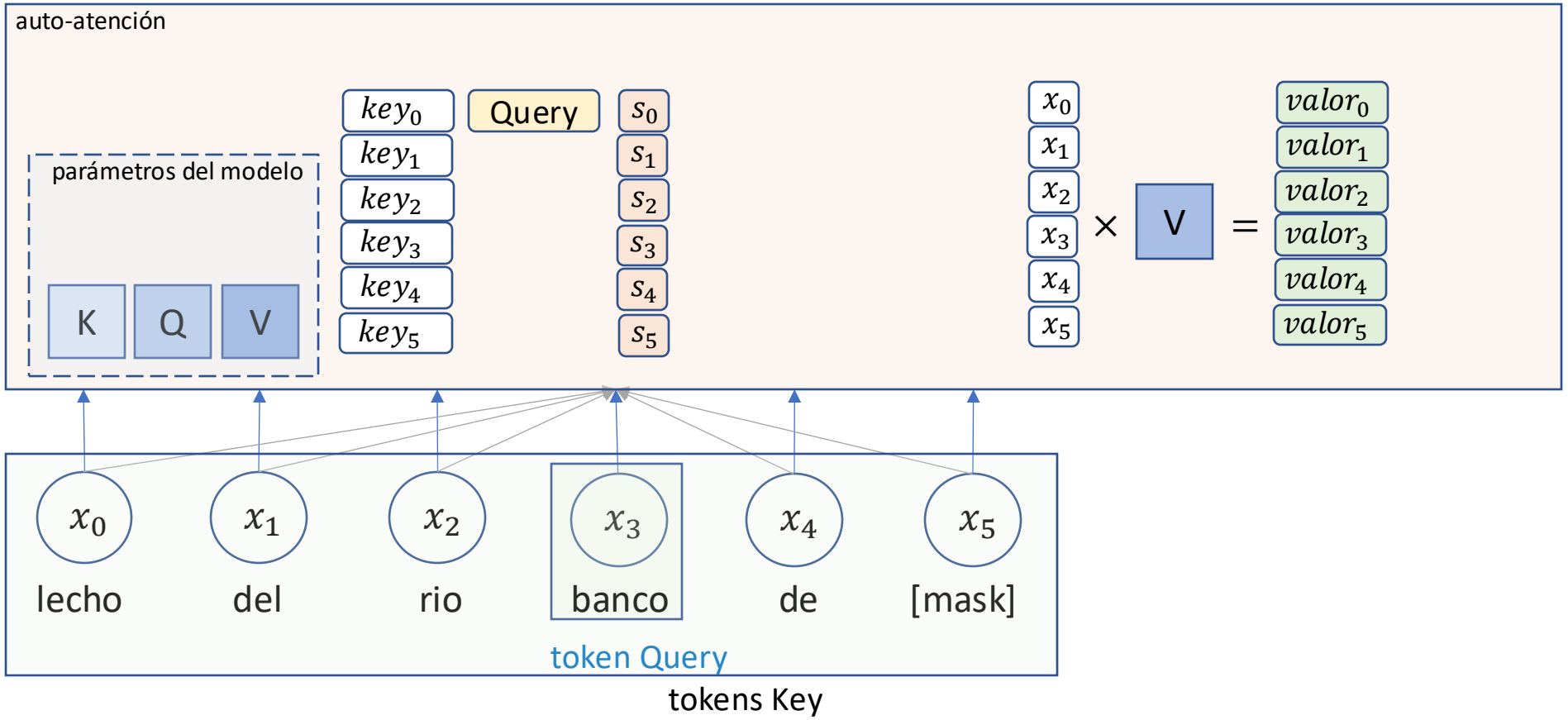


Métrica de similaridad

También conocido como el Coseno de Similaridad

Transformers

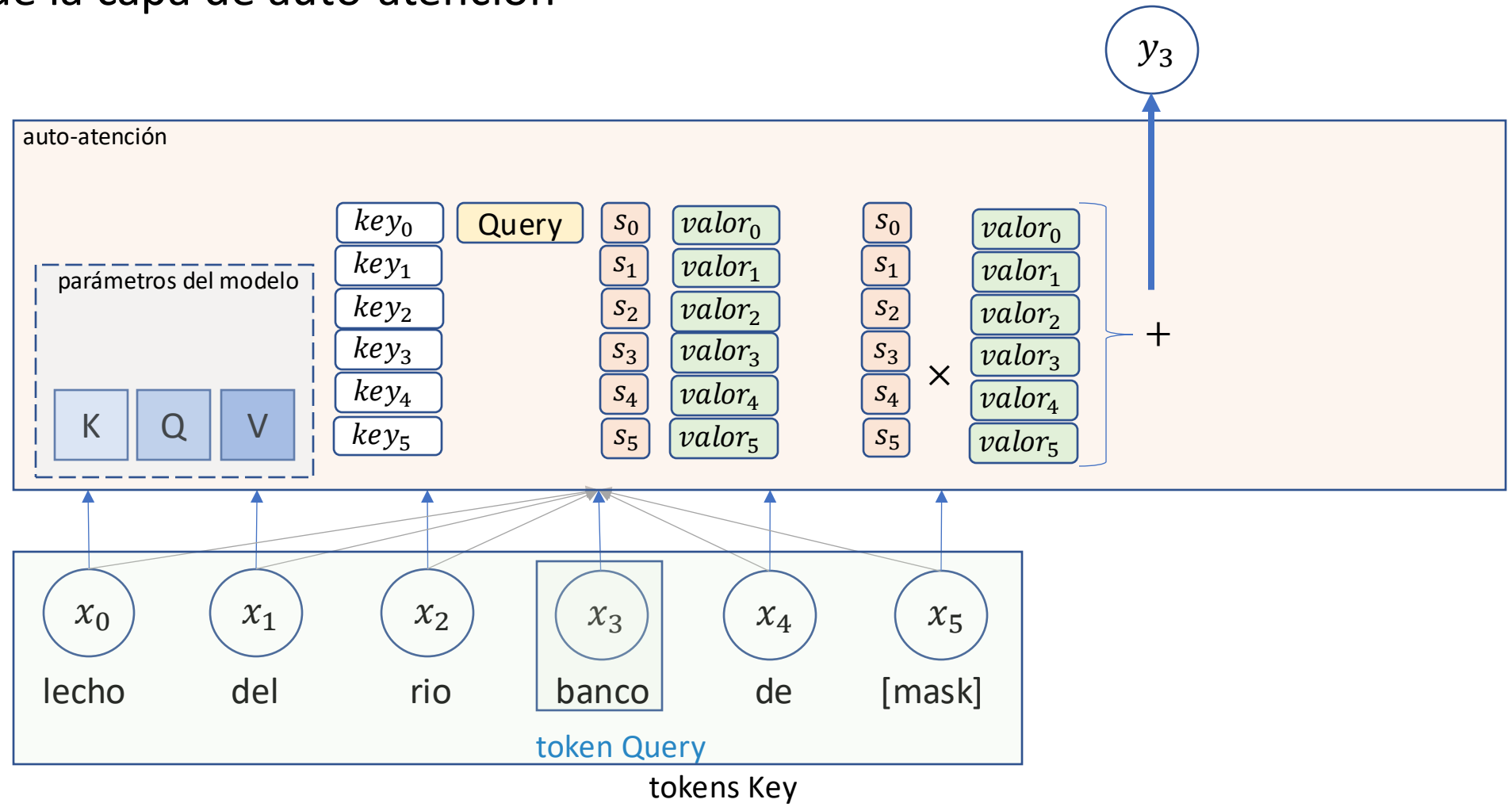
- Las Valores



Embeddings no-Contextual X_i

Transformers

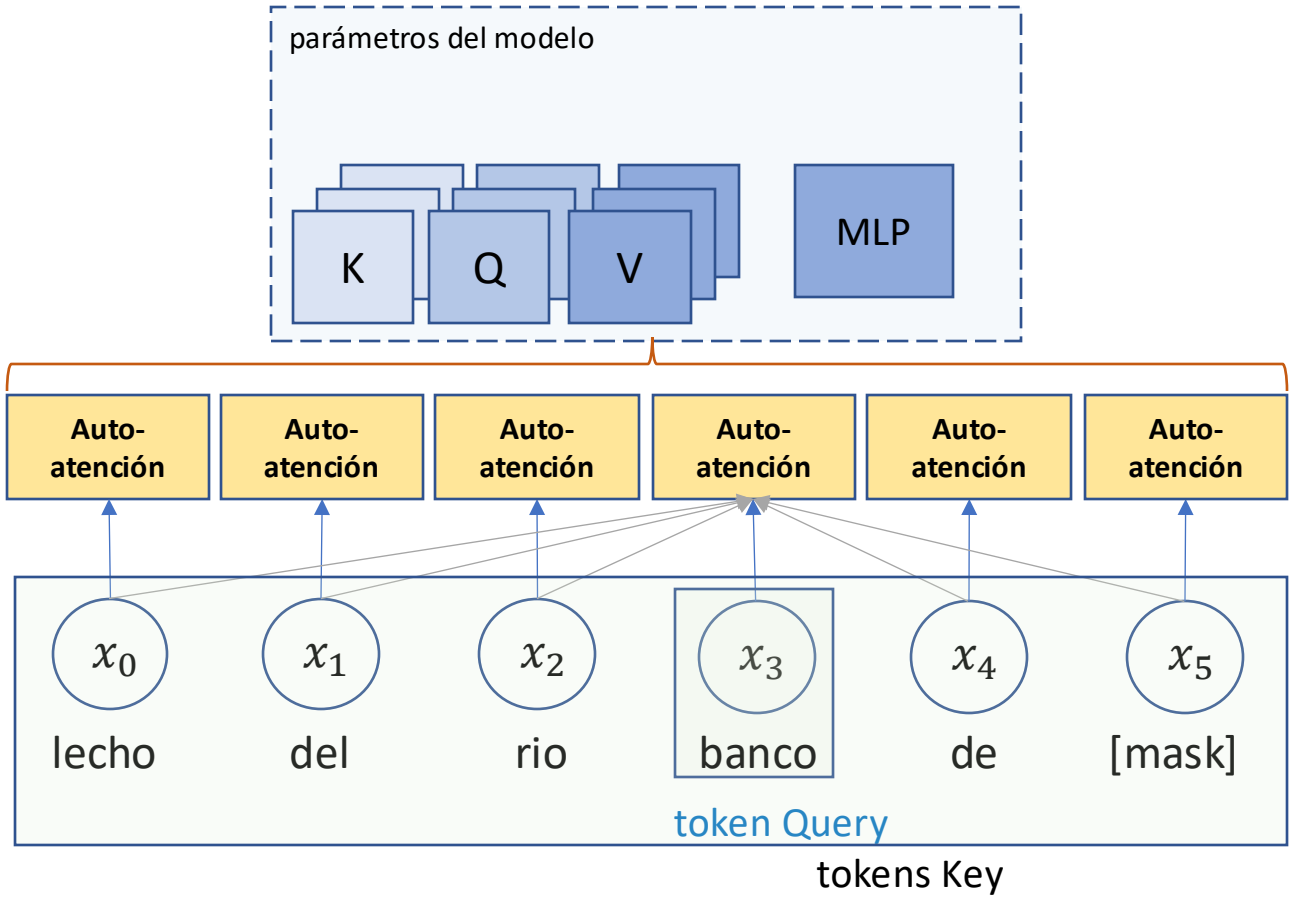
- La Salida de la capa de auto-atención



Embeddings no-Contextual X_i

Transformers

- Auto-atención con múltiples cabezas



Embeddings no-Contextual X_i

Transformers

- El codificador - Transformer

